Activity 8: Management Capstone: Build the Apex ESD Control Plan

Objective: Integrate the complete course into a management-ready ESD control plan and defend it under customer-audit-style questioning.

Scenario

Apex Microsystems is opening a mixed lab/factory line for sensitive industrial controllers. Management needs an auditable ESD control plan before customer qualification, covering controls, verification, competence, incidents and improvement.

Roles

- ESD programme manager
- operations manager
- quality manager
- customer auditor

Detailed procedure

1. Define scope, ESDS products, process boundaries, applicable customer/standard criteria and programme objectives.
2. Assign programme coordinator, technical owners, area owners, operators and management-review responsibilities.
3. Create the process risk map and select controls using elimination, grounding/bonding, ionisation, shielding and administrative layers.
4. Define EPA requirements for people, workstation, equipment, materials, packaging, logistics and field service.
5. Build the qualification register for each control item and required evidence.
6. Build the compliance-verification matrix with item, method, limit/source, frequency, owner, record and response.
7. Create the role-based training and competency matrix with initial, observed and refresher evidence.
8. Define incident reporting, containment, investigation, CAPA and effectiveness verification.
9. Select leading, process, lagging and learning metrics with targets and review frequency.
10. Run a tabletop scenario for a failed personnel tester and an ioniser drift; trace every decision and record.
11. Conduct a customer-auditor challenge using the scenario questions.
12. Revise, approve and issue the plan with a 90-day implementation roadmap.

Questions

- How does the plan demonstrate control, verification and improvement?
- Which records prove the control programme remains effective?
- What management decision is triggered by deteriorating leading indicators?

Evidence to submit

- ESD control plan
- Qualification and verification matrices
- Competency matrix
- Metrics dashboard and 90-day roadmap

Acceptance criteria

- [ ] All learning outcomes are traceable
- [ ] Controls have owners and evidence
- [ ] Failure response and CAPA are closed-loop
- [ ] Plan uses current controlled standards for exact requirements

Debrief

Explain which evidence changed your first assumption, which control is most reliable, and what follow-up measurement is needed.